1. **12. Identify the constraints, assumptions simplifications in a mathematical model**
   - The problem explicitly lists several assumptions ("he makes the following assumptions"), which must be recognized in modeling (e.g., each pair raises one chick, 20% of total penguins die at year-end).
2. **1. Select a mathematical description or a representation that describes a problem**
   - Simplifies real-world population dynamics into two stages (breeding + death) and chooses appropriate mathematical expressions to represent changes in numbers.
3. **4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
   - Information is given in text; one must interpret it and build a model of the chronological sequence and quantitative relationships.
4. **9. Translate a problem into a standard mathematical representation or algorithm**
   - Can be translated into a linear formula: year-end count = [(initial adult count) + (number of pairs × chicks per pair)] × (1 − mortality rate).
5. **8. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures**
   - The problem is essentially a piecewise linear growth/death model, similar to simple population dynamics models.
6. **7. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis**
   - Decomposes the annual process into two phases: breeding (addition) → death (multiplication), which simplifies the problem.

1. **12. Identify the constraints, assumptions simplifications in a mathematical model**
   - 题目直接列出了几条假设（“he makes the following assumptions”），建模时必须识别这些条件（如：每对养 1 只雏鸟，年底总数量 20% 死亡）。
2. **1. Select a mathematical description or a representation that describes a problem**
   - 将现实中的种群动态简化为两个阶段（繁殖 + 死亡），并选择适当的数学表达式表示数量变化。
3. **4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
   - 题目信息以文字给出，需要解读并建立时间顺序和数量关系模型。
4. **9. Translate a problem into a standard mathematical representation or algorithm**
   - 可以翻译成线性公式：年末数 = [(年初成年数) + (对数 × 每对雏鸟数)] × (1 - 死亡率)。
5. **8. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures**
   - 该问题本质上是分段线性增长/死亡模型，与人口动态的简单模型类似。
6. **7. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis**
   - 将一年的过程分解为两个阶段：繁殖（加法）→ 死亡（乘法），这种分解简化了问题。